

# Paper 4.25 - Toolkit for Boundary Repair

CQE-paper-04.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

## Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

## Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-04.25.md

## Paper 4.25 - Toolkit for Boundary Repair

### Purpose

Paper 4.25 describes the review tools for boundary repair. The tools help a reader construct and inspect repair rows. They do not add claims beyond the Paper 4 theorem.

### Review Objects

The toolkit works with:

|                       |                                           |
|-----------------------|-------------------------------------------|
| correction state      | = Paper 2 residue state                   |
| axis_sheet coordinate | = Paper 3 registration                    |
| repair row            | = typed boundary constraint               |
| next legal route      | = route allowed to consume the constraint |

### Digital Toolkit

Primary executable files:

```
production/formal-papers/CQE-paper-04/verify_boundary_repair.py
production/formal-papers/CQE-paper-04/boundary_repair_receipt.json
```

Additional package evidence named by the dyad review:

```
D:\CQE_CMPLX\CMPLX-R30-main\PROOF\src\lattice_forge\binary_boundary_adapter.py
D:\CQE_CMPLX\CMPLX-R30-main\PROOF\src\lattice_forge\transport_obligations.py
```

The verifier checks that Paper 2 correction states are consumed, Paper 3 coordinates are preserved, required fields are present, constraints are not proofs, next legal routes are nonempty, repair is idempotent, and untyped failures are rejected.

### Analog Toolkit

A physical reconstruction requires:

- one correction row,
- one coordinate lookup,
- one repair-row template,
- one next-route label,
- one receipt line.

Procedure:

```
copy the Paper 2 correction state
copy the Paper 3 axis/sheet coordinate
write the reason: correction fired
write status: constraint
write at least one next legal route
write source paper and target paper
repeat the repair operation and confirm the row does not change
```

### Hidden-Guess Diagnostics

When training mode is enabled, diagnostic review should ask for a prediction before revealing the receipt:

What fields are required for a repaired boundary row?  
Does a repaired row count as proof?  
Which two states produce repair rows?  
Is {"status":"failed"} repairable?

Expected answers:

state, axis\_sheet, reason, status, next\_legal\_routes, source\_paper, target\_paper  
no  
(0,1,0) and (1,1,0)  
no

## Boundary

Paper 4.25 is a toolkit supplement. If a new adapter, route, or domain example is found, it must pass Paper 4.50's claim contract before it changes the scientific paper.